

- 9 C By conservation of energy,  
total energy at B = total energy at A  
 $mgr = \frac{1}{2}mv_B^2$   
 $\Rightarrow v_B = \sqrt{2gL}$

At B,

$$\text{resultant force} = \frac{mv^2}{L} = T - mg$$

$$\Rightarrow \frac{m(2gL)}{L} = T - mg$$

$$2mg = T - mg$$

$$T = 3mg$$